

What Open-Loop Hides and Closed-Loop Reveals: A Study of Weight Quantization, Generation Coherence, and Edge Deployment for Autonomous-Driving Vision–Language–Action Models

Justin Williams, Kishor Datta Gupta, Roy George, Mrinmoy Sarkar
Clark Atlanta University
justinwilliamstech693@gmail.com

Abstract—We present a comprehensive study of post-training weight quantization for autonomous-driving vision–language–action (VLA) models, using OpenDriveVLA-0.5B (a Qwen2.5-0.5B language/action head on UniAD stage-1 track+map perception). We make four contributions. First, we show that the standard open-loop nuScenes L2 metric is structurally blind to quantization: INT8/INT4 weight quantization is lossless (0.33 m), yet a controlled ego-ablation (zeroing the ego-state/history prompt) inflates L2 to 6.38 m (19 $\times$), demonstrating the metric is $\approx$ ego-motion extrapolation. Second, we build a native (Docker-free) closed-loop harness around the NeuroNCAP neural-rendering simulator and the OpenDriveVLA policy, including a synthesized CAN-bus expansion and an in-place re-quantization service for fast technique sweeps. Third, we report the central empirical result, replicated across 14 real-camera NeuroNCAP scenes (16 scenario instances): under the deployment-domain shift of neural rendering, naive per-output-channel INT4 collapses trajectory-text generation ($\leq 15\%$ well-formed on 15/16 scenarios, mean 3.7%; planner frozen), whereas group-wise INT4 (group 128, ≈ 0.1 extra bits/weight) recovers the bulk of FP16-level generation coherence (mean 73.5% vs 76.0%; $\geq 85\%$ of FP16 on 13/16 scenarios). The collapse is a quantization $\times$ domain-shift interaction governed by scale granularity, not bit-width, since the same per-channel INT4 is lossless in-distribution. Fourth, we provide component memory benchmarks and an edge-deployment analysis (Jetson Orin, Raspberry Pi) showing the planner is trivial for the edge and perception is the bottleneck. We are explicit throughout about scope: group-wise recovery is partial on 3 scenarios, and several scenes are failed by every configuration including FP16—so we lead with the coherence metric and identify exactly what additional evidence a fleet-wide claim requires.

Index Terms—vision–language–action models, autonomous driving, post-training quantization, group-wise INT4, closed-loop evaluation, generation coherence, edge deployment, Jetson Orin

I. Introduction

On-board inference for end-to-end driving stacks motivates compressing the increasingly large language backbones of driving VLAs. The prevailing evidence—near-lossless 4-bit weight quantization on open-loop nuScenes planning—suggests this is essentially free. We argue, and

demonstrate, that this evidence is structurally unable to support a safety claim, and that closed-loop evaluation tells a different and more actionable story. We organize the paper around four questions: (Q1) Can open-loop L2 see quantization damage? (Q2) Does quantization affect closed-loop behavior, where the logged ego trajectory cannot be coasted on? (Q3) If so, what property of the quantizer governs the effect, and what is the fix? (Q4) What does this imply for edge deployment?

II. Related Work

Driving VLAs and UniAD. UniAD introduced a planning-oriented, query-based perception stack (multi-camera ResNet backbone, temporal BEVFormer, track/map/motion/occupancy heads). OpenDriveVLA reuses its track+map perception as a vision tower for a Qwen language/action head that emits trajectories as text. Open-loop critique. Prior analyses observed that open-loop nuScenes planning is dominated by ego state; we quantify this for a quantization context. Closed-loop neural simulation. NeuroNCAP couples a NeuRAD neural renderer to a safety-test protocol, enabling photorealistic closed-loop evaluation on nuScenes scenes. LLM weight quantization. Round-to-nearest, GPTQ, and AWQ, with per-tensor/per-channel/group-wise scale granularity; group-wise is standard for INT4 because it bounds the influence of weight outliers on the shared scale.

III. System and Method

A. Model and quantization

OpenDriveVLA-0.5B: Qwen2.5-0.5B (hidden 896, 24 layers) on UniAD track+map perception. We apply weight-only post-training quantization to the 168 Linear layers of the Qwen backbone (357.8M params), dequantizing to the compute dtype (fake-quant), so the inference path is unchanged and only the language head’s W_b accuracy is altered. Perception, the scene/track/map projectors, token embeddings, and the output head are

left in floating point. We compare scale granularities (per-output-channel; group sizes 128 and 64) and symmetric vs. asymmetric round-to-nearest. An activation-aware (AWQ) path exists in the code but is excluded from claims pending a fix to its low-bit behavior.

B. Open-loop evaluation

We evaluate on the 162 nuScenes-mini $\cap$ validation samples under the ST-P3 and UniAD L2 protocols. The ego-ablation zeroes the ego-state feature vector and the 2s history trajectory in the planner prompt, leaving perception untouched.

C. Closed-loop harness

We run three native processes communicating over local-host HTTP (no Docker): (i) the NeuroNCAP orchestrator (kinematic vehicle model, collision check, scoring); (ii) the NeuRAD renderer serving six photorealistic camera views at the ego’s driven pose; (iii) an OpenDriveVLA model node. The model node reconstructs the UniAD perception input from rendered frames each step—image normalization/padding, lidar-to-image projection, an 18-vector CAN-bus signal, and the ego-state text prompt—and exposes an in-place re-quantization endpoint so techniques are swept on the loaded model without reloading, and an activation-capture endpoint for calibration. Because the gated nuScenes CAN-bus expansion was unavailable, we synthesized scene-0103’s CAN-bus pose/steering messages from the dense lidar ego-pose track (used only for priming/initial velocity; the simulator emulates CAN-bus from driven poses thereafter). Engineering details, including renderer fixes for the nuScenes dataparser and a guard for missing map ground truth on rendered inputs, are in the code release.

D. Metric: generation coherence

Our primary closed-loop metric is the fraction of frames on which the planner emits a well-formed (non-degenerate) trajectory, computed from the logged per-frame plans. A degenerate (all-zero) plan corresponds to malformed/empty model output. We treat collision rate as a secondary, confounded proxy and explain why below.

IV. Results

A. Q1: Open-loop L2 is blind

Weight quantization is lossless to INT4. The INT3/INT2 degradation is generation collapse (malformed text), not graded planning. The ego-ablation shows the metric is $\approx$ ego-extrapolation: with ego state removed the plan collapses $19\times$ while outputs stay well-formed. Open-loop L2 thus cannot reflect perception/planning, nor quantization damage to them.

a) The blindness persists on the deployment target (Jetson AGX Orin): We replayed the captured planner inputs on a real AGX Orin 64 GB (JetPack 6, torch 2.8) and re-scored open-loop L2 (40-sample mini subset, ST-P3). The Orin reproduces the A100 to ≈ 1 cm mean trajectory difference (FP16), and open-loop L2 is essentially unchanged by quantization on-device: FP16 0.196/0.453/0.870 m vs group-INT4 0.206/0.475/0.897 m vs per-channel-INT4 0.204/0.467/0.882 m at 1/2/3 s. Crucially, the per-sample W4 trajectories already diverge from FP16 by 14–24 cm (mean) on the Orin while the aggregate L2 moves < 3 cm—the metric averages away a real quantization perturbation. Open-loop blindness is therefore a property of the metric, reproduced on the actual edge hardware, not an artifact of the evaluation server.

B. Q2–Q3: Closed-loop reveals collapse; granularity is the fix

Across all 16 scenario instances, per-channel INT4 collapses generation—degenerate (all-zero/ malformed) plans on essentially every frame, $\leq 15\%$ well-formed on 15/16 scenarios (mean 3.7%)—while group-wise INT4 recovers the bulk of FP16’s coherence (mean 73.5% vs 76.0%), reaching $\geq 85\%$ of FP16 on 13/16 scenarios. The same per-channel INT4 is lossless in-distribution (Table 1), so the failure is a quantization $\times$ domain-shift interaction. We are careful not to over-claim equivalence: group-wise only partially recovers on 3 scenarios (0099, 0101, 0796).

a) Why collision/NCAP is a confounded, scene-dependent proxy.: On scene-0796 every configuration—including FP16—collides on 10/10 seeds: the base policy cannot clear that scenario under our rendering, so collisions cannot isolate the quantization effect there even though coherence still separates the configs cleanly ($\approx 58\text{--}72\%$ vs 8.6%). On 0103 collisions do track collapse (per-channel 4/10 vs 0/10). Because the outcome is dominated by scene difficulty and a degenerate stay-in-place fallback, we lead with generation coherence.

b) Mechanism.: An INT4 group shares scale $s = \max |w|/7$; a single large weight inflates s and coarsens the grid for the small, signal-carrying weights it shares the scale with. Per-channel shares one scale across an entire row; group-wise confines each outlier to a 128-weight block. Under distribution shift the planner’s hidden states move off the calibrated regime, and the coarsened per-channel grid tips greedy decoding into malformed trajectory text; the finer group-wise grid preserves coherence. The cost is 16/ g extra bits/weight (0.125 at $g=128$).

C. Q4: Memory benchmarks and edge deployment

The robust recipe (group-wise INT4) is ≈ 0.26 GB and the whole node fits in 3–5 GB, so the planner is trivial for a Jetson AGX Orin (32/64 GB). The obstacle is perception throughput: UniAD’s six-camera ResNet-101 + deformable-attention BEVFormer runs FP32 with

TABLE I
Open-loop nuScenes-mini ($n=162$), ST-P3 L2 (m).

Setting	L2 (m)	well-formed
FP16	0.33	100%
INT8 (per-channel)	0.33	100%
INT4 (per-channel)	0.33	100%
INT3	1.91	92%
INT2	6.45	0%
FP16, ego-state+history zeroed	6.38 (19 $\times$)	100%

TABLE II
NeuroNCAP closed-loop generation coherence (well-formed predicted-frame rate, %), real-camera NeuRAD rendering, 14 scenes / 16 scenario instances, 10 paired seeds each.

scene / scenario	FP16	INT4 per-channel	INT4 group-128
0099 stationary	60.6	0.0	30.0
0101 stationary	75.0	14.3	46.7
0103 frontal	74.3	0.0	72.9
0106 frontal	66.9	0.0	78.7
0108 side	85.7	0.0	91.6
0108 stationary	72.4	0.0	74.7
0110 frontal	70.7	0.0	74.3
0278 side	75.3	0.0	67.9
0278 stationary	79.7	0.0	83.5
0331 stationary	83.2	5.0	97.5
0346 frontal	83.1	15.6	86.9
0783 stationary	72.4	0.0	73.1
0796 stationary	72.4	8.6	57.9
0921 side	90.0	14.2	90.4
0923 frontal	74.6	0.8	72.1
0966 stationary	79.4	0.0	78.3
mean	76.0	3.7	73.5

TABLE III
Backbone weight footprint/overhead (357.8 M params); effective bits include per-group FP16 scales. Full inference node measured at 3–5 GB (perception-dominated).

Precision	eff. bits/weight	backbone weights
FP16	16.0	$\approx$ 1.0 GB
INT8 per-channel	8.06	$\approx$ 0.50 GB
INT4 per-channel	4.06	$\approx$ 0.25 GB
INT4 group-128	4.125	$\approx$ 0.26 GB

custom CUDA kernels, far from the ~ 10 Hz a vehicle needs without a TensorRT-INT8 port (ARM64/ JetPack rebuilds, deformable-attention plugins). On a Raspberry Pi the gap is categorical: no CUDA and no CPU path for the custom ops, so perception must be replaced, not ported, while the INT4 planner is borderline-feasible via llama.cpp ($\sim$ seconds/plan, estimated). Edge effort should target perception; the planner is a solved sub-problem.

V. Discussion

The study composes into a single message: open-loop L2 hides quantization damage because it is ego-extrapolation; closed-loop reveals it because perception and planning are on the critical path; and the revealed damage is a granularity artifact with a near-free fix. The methodological recommendation (do not certify quantized driving policies on open-loop L2) and the engineering recommendation

(default to group-wise INT4 for the planner) are both actionable today.

VI. Threats to Validity

Scene coverage. Closed-loop results span all 14 NeuRAD-reconstructed NeuroNCAP scenes (16 scenario instances across frontal/side/stationary); this is the full released closed-loop benchmark, not all of nuScenes. Renderer domain gap. The rendered inputs are out-of-distribution relative to real nuScenes; on several scenarios even FP16 fails the collision test (10/10), so absolute NCAP/collision numbers are not comparable to in-distribution operation and our claims are explicitly relative (group $\approx$ FP16 $\gg$ per-channel on coherence). Partial recovery. Group-wise INT4 ties or beats FP16 coherence on 13/16 scenarios but only partially recovers on 3 (0099, 0101, 0796); we do not claim exact equivalence.

Metric proxy. Collision/NCAP is confounded and scene-dependent; we lead with coherence. INT3/INT2/AWQ. Lower-bit and activation-aware variants were characterized only in an earlier single-scene setup and are not re-reported here pending a real-camera re-run. AWQ. The activation-aware low-bit path is buggy and excluded. Fake-quant. Weights are dequantized for inference, so a packed-INT4 speedup is not yet realized (needs a custom kernel); the static footprint, however, is measured on-device. On-Orin measurement. The planner was run on a Jetson AGX Orin 64 GB (JetPack 6, torch 2.8): 2.88 GB runtime peak, 14.9 tok/s (1024-ctx, 60-token decode, eager, default clocks), group-wise INT4 packing the 357.8 M-param planner $0.72 \rightarrow 0.19$ GB ($3.9\times$); latency is flat across precisions under fake-quant. Perception throughput on Orin remains a device-class estimate (no TRT port).

VII. Reproducibility

The code release includes the open-loop evaluation and quantization sweep scripts, the ego-ablation toggle, the three-process closed-loop harness with launch instructions, the in-place re-quantization service, the per-seed sweep drivers, the CAN-bus synthesis, and a video renderer for the closed-loop rollouts. All numbers in this paper are produced by those scripts; estimates are marked as such.

VIII. Conclusion and Future Work

Open-loop L2 is blind to quantization in driving VLAs ($19\times$ ego-ablation); closed-loop reveals that per-channel INT4 collapses generation under deployment-domain shift while group-wise INT4 (≈ 0.1 extra bits/weight) matches FP16. Future work: more scenes and a stronger renderer to test generality and disentangle the domain-gap confound; a direct malformed-output-rate protocol; a corrected activation-aware quantizer; and on-Orin throughput with a TensorRT-INT8 perception port.

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